This hackathon project is about building an AI-powered industrial “co-pilot” that can design, cost, simulate, and manage a factory or production line end-to-end, from technical specs to digital twin and business plan.

Global idea of RIBIF

The RIBIF system (name they give to the solution) is basically an intelligent assistant for industrial investment projects. It should help an industrial company go from a technical datasheet of machines/materials to a complete virtual factory: layout, suppliers, negotiations, costs, financial plan, and digital twin with KPIs like OEE/TRS.
The final deliverable is one integrated system (web app / platform) that chains 9 mandatory modules together in a coherent workflow.

Module 1 – Tech sheets & needs

This module ingests technical datasheets (PDF, DOC, etc.) and automatically extracts:

- Key specifications of machines/materials and process constraints.
- A structured understanding of needs: capacities, required utilities, safety, etc., plus a first classification of materials, machines, and processes into families.

The idea is to turn messy human documents into clean structured data that all other modules can use (for layout, sourcing, costing, etc.).

Module 2 – 2D/3D factory layout

From the specs and needs, this module generates:

- 2D plans in DWG format (CAD-style) and 3D models in IFC format, representing the factory or workshop.
- A complete layout: spaces, flows (materials, people), zoning (production, storage, utilities, offices), respecting basic industrial design logic.

This is where BIM / CAD / procedural generation / AI-assisted layout come in, plus possibly external 3D tools (Blender, etc.).

Module 3 – HD video rendering

This part turns the 3D model into:

- Photorealistic interior and exterior renders of the plant.

- A neutral-branding HD video (45–60 seconds) “tour” of the site (for example using tools like Runway, Luma, Meshy or traditional rendering).

It is mainly for visualization and “wow factor” for investors, management, or non-technical stakeholders.

Module 4 – Automated supplier sourcing

Given the list of machines, equipment, and materials, this module must:

- Identify potential suppliers worldwide and scrape/collect data on prices, lead times, quality ratings, etc.
- Build an automatic comparison table (per item or family) with scoring or ranking.

Think: semi-automated procurement search engine with structured comparison.

Module 5 – AI negotiation (text + voice)

This module simulates or supports negotiation with suppliers:

- Text agent: writes emails, counter-offers, evaluates Incoterms and conditions, and iterates to get a better deal.
- Voice agent: able to converse in English with a supplier, and provide transcripts and a “best offer” summary at the end.

Here you show off LLM + voice (TTS/STT) + some negotiation logic.

Module 6 – TCO (Total Cost of Ownership)

From previous modules, this one computes the real industrial cost of the solution:

- Purchase, installation, energy, maintenance, consumables, etc., over a defined horizon.
- Outputs a realistic TCO per machine, line, and possibly per product, reflecting the “true” industrial cost beyond just CapEx.

This is where optimization, costing models, and maybe scenario comparison appear.

Module 7 – Automatic business plan

Using TCO and production assumptions, this module automatically builds:

- A 5-year financial model in Excel: CapEx, OpEx, pricing hypotheses, revenues, ROI, payback, etc.

- An auto-generated pitch deck summarizing the project for investors or management.

Essentially, you turn the technical/operational design into a financial story with numbers.

Module 8 – Digital twin & “digital brain”

Here the factory becomes a living digital twin:

- Numerical twin of the line/plant with AI for scheduling, quality control, and predictive maintenance.
- A dashboard with industrial KPIs such as OEE/TRS, throughput, availability, quality, and other performance indicators.

The “digital brain” is the set of AI models making decisions or recommendations to optimize operations.

Module 9 – Automatic technical catalog

Finally, the system outputs:

- A technical catalog of machine families, product families, process chains, and logistics chain, all linked to the digital twin.
- A PDF and web version that document the proposed industrial solution and its main parameters.

This is like a structured, reusable knowledge base of the designed factory and its components.

If you want, next step can be to map your skills to each module and decide a realistic MVP cut (e.g., strong prototype for 3–4 modules with light stubs for the others).